

No. of Functional Features Included in the Solution

Date	1 October 2026
Team ID	SWTID-2026-2533
Project Name	EduGenie Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Functional Features Overview:

This document lists every functional feature in the EduGenie solution, with its purpose, the module that implements it and its current status.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contrib.
1	Question Answering	Answers academic questions (for example, about oceans) in a short, clear form using Gemini.	qna.py, /qa	Done	High
2	Concept Explanation	Explains a topic in simple words for a school student using the local LaMini-Flan-T5 model, with Gemini as fallback.	explanation_module.py, /explain	Done	High
3	AI Quiz Generation	Creates 3 multiple-choice questions (4 options each) for a topic or passage, returned as validated JSON.	quiz_module.py, /quiz	Done	High
4	Text Summarization	Condenses long passages into short, simple-language summaries for quick revision.	summary_module.py, /summarize	Done	Medium
5	Learning Path	Builds a beginner, intermediate and advanced roadmap with timelines and resources (for example, SQL).	learning_path.py, /learn/recommendations	Done	High
6	Responsive Web Interface	One page with a task dropdown, a text box and a submit button that works on phones and desktops.	templates/index.html, static/style.css	Done	Medium
7	Quiz Feedback and Formatted Results	Shows Correct / Incorrect with the right answer for each quiz question, and renders headings, bold text and bullet lists safely.	index.html (JavaScript)	Done	Medium
8	Error Handling and Input Validation	Rejects empty input and shows clear messages when the AI service or network fails.	main.py (run(), error())	Done	Medium
9	Secure Configuration and Deployment	Keeps the API key and model name in environment variables and serves the app online on Vercel.	gemini_client.py, .env, Vercel	Done	Medium

Feature Summary:

Metric	Count / Value
Total Features Planned	9
Total Features Implemented	9
Core / Must-Have Features	6
Additional / Nice-to-Have Features	3
Features Tested & Verified	9

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Responsive single page with task dropdown, quiz answer feedback and formatted results (HTML, CSS, JavaScript).
2	Backend / Logic	3	Local concept-explanation pipeline, quiz JSON cleaning and validation, and shared error handling in FastAPI.
3	Database / Storage	0	Not applicable: EduGenie is stateless and stores no user data. Progress storage is future scope.
4	API / Integration	3	Google Gemini connection for Q&A, summaries and learning paths through the Google GenAI SDK.
5	Security / Configuration	1	API key and model name in environment variables, .env excluded from Git, and safe escaping of AI text in the page.